

Contacts

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Languages

Italian (Native)

English (Professional)

French (Basic)

Technical Skills

Machine learning & modeling

PyTorch/Geometric,
Scikit-Learn, TensorFlow,
Optuna, Gensim, SHAP

Generative AI & LLMs

LangChain, LoRa

Programming

Python, C++, R, Julia,
Java, SQL

Data processing & analytics

Pandas/Geopandas,
Numpy/Scipy, NetworkX,
PySpark

Software & tools

Git, Jupyter, VSCode,
LateX, Microsoft Office

Certifications (NVIDIA Deep Learning Institute)

RAG Agents with LLMs

Transformer-based
Applications

CUDA C++ with Multiple
GPUs

Fundamentals of CUDA
Python and CUDA C/C++

Experience

2023–Now

Postdoctoral Researcher

Department of Computer Science, University of Pisa, IT

– KDD Lab member, mentored by Profs. Dino Pedreschi and Riccardo Guidotti. ERC XAI project team in 2023-2025. Co-writer of the ERC Proof-of-Concept ILLUME-4-Science and WP-leader in 2025-2026. TANGO and FAIR collaborator.

– Developed novel ante-hoc and post-hoc interpretable tabular data methods (counterfactuals, rule-based, and feature-importance explanations); designed self-explainable neural networks for graphs.

Oct.22-Feb.23

Research Intern

CENTAI Institute, Turin, IT

– Internship within the Responsible AI group led by Principal Researcher Andr  Panisson.
– Developed retrofitting methods for obtaining node embeddings with interpretable features.

Jun.22-Sep.22

Research Intern

L3S Research Center, Hannover, DE & Delft University of Technology, NL

– Winner of the SoBigData++ TNA Grant for "Social Impact of AI and explainable ML", in collaboration with Professors Avishek Anand and Megha Khosla.

– Designed mechanistic explainers for opaque embeddings, with benchmarking of existing embedding methods for graphs.

2017–2022

Data Science Researcher

ISI Foundation, Turin, IT

– Lagrange Applied Research Scholarship owner (2017-2018); PhD candidate at University of Bologna, based at ISI Foundation (2018–2022).

– Developed node embedding algorithms for dynamic graphs and hypergraphs, with applications in link prediction for social network analysis and epidemiological forecasting.
– Inferred human behavior from CDRs to characterize gender-based urban mobility; Developed urban-level income predictors from satellite imagery and FB Ads attributes.

Education

2018–2023

PhD in Data Science and Computation [Dissertation]

University of Bologna, IT

Thesis: Learning Representations for Graph-Structured Socio-Technical Systems (node embeddings, dynamic networks, link prediction). Supervisors: Ciro Cattuto, Andr  Panisson.

2014–2017

MSc in Physics of Complex Systems [Grade: 110/110 cum laude]

University of Turin, IT

Thesis: Statistical Analysis and Modelling of Users' Exploratory Behaviour in Vector Embedding Spaces (word embeddings, large-scale text processing, user modeling).

2011–2014

BSc in Physics [Grade: 110/110]

University of Turin, IT

Thesis: "Quantum Information and Teleportation on Quantum Memories".

Honors & awards

2026

KDD Best Reviewer - Awarded among the best reviewers in the Research Track of KDD 2026.

2026

ERC Proof of Concept - Co-writer and WP leader of EU-granted ILLUME-4-Science (18-month project for tech transfer translating XAI research prototypes into applied ML tools) [link].

2025

ICDM Outstanding Poster Award - Awarded for the best poster at ICDM 2025.

2022

SoBigData++'s Transnational Access - Grant winner in the track "Social Impact of AI and Explainable ML" (2-month scholarship for funding research visits) [blog].

2017

Lagrange Fellowship in Data Science for Social Impact - Grant winner (1-year applied research scholarship) [link].

Selected publications

KDD 2026

Explainable AI for Cancer Drug Response Prediction: Beyond Univariate Feature Attributions [link](#), [code](#)

ICDM 2025

Post-Hoc Explainability for Tabular Data with Interpretable Meta-Encoding [link](#), [code](#)

TMLR 2025

Disentangled and Self-Explainable Node Representation Learning [link](#), [code](#)

TKDE 2024

DINE: Dimensional Interpretability of Node Embeddings [link](#), [code](#)

LoG 2022

Higher-order Link Prediction and Reconstruction from Simplicial Complex Embeddings [link](#), [code](#)

EPJ DS 2022

Time-varying Graph Representation Learning via Higher-order Skip-gram with Negative Sampling [link](#)

Selected talks

- Nov, 2025 **ICDM 2025** - "Explanations go linear: post-hoc explainability for tabular data with interpretable meta-encoding". Washington DC, USA
- Mar, 2025 **Second FAIR Workshop on Human-Centered AI** - "Explainable AI for synergistic human-AI collaboration with ILLUME". Pisa, IT
- Oct, 2022 **Conference on Complex Systems** - "Simplicial complex embeddings for higher-order link prediction and reconstruction". Palma de Mallorca, ES
- Sep, 2020 **NetSci Satellite on Machine Learning in Network Science** - "Time-varying graph representation learning via higher-order skip-gram with negative sampling". Online
- Oct, 2018 **Workshop on Complexity and the City** - "Predicting city poverty using satellite data". Turin, IT

Events organization

- May, 2026 **(X)AI4Science ERC PoC XAI Project Kick-off** Co-organizer. Pisa, IT
- Jun, 2025 **International Conference on Hybrid Human-Artificial Intelligence** Local organizer. Pisa, IT
- Jul, 2024 **First XAI Hackathon** Co-organizer and designer of a real-world XAI challenge related to churn prediction. Pisa, IT

Professional development

- Sep, 2025 **M2L Summer School** - Advanced training on Generative and Agentic AI, organized by Google DeepMind (acceptance rate $\sim 17\%$). Split, HR
- Jul, 2025 **LOGML Summer School** - Challenged in group project on knowledge injection and LoRA fine-tuning for biomedical LLMs (acceptance rate $\sim 30\%$). London, UK
- Apr, 2024 **TANGO Codecamp** - Training sessions and challenges on software engineering practices (APIs, UML, CI/CD, MLOps). Barcelona, ES
- Jun, 2019 **Workshop Complexity72h** - Publication-oriented group challenge studying triadic motif analysis in the mergers and acquisitions economic network. Lucca, IT